Tampere University

# BBT.HTI.501 Processing of Biosignals Assignment 3

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# Signals

Parent ECG

(a)

Parent + Fetus ECG

(b)

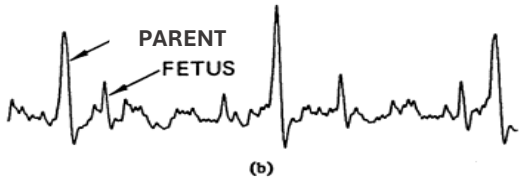
Fetus ECG ??

# Adaptive filtering

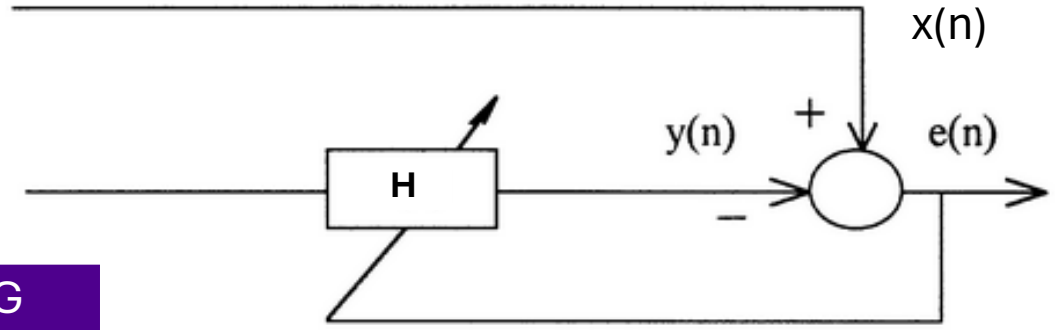
- Adaptive filtering adapts to produce a better filtering result, when the algorithm is run several times
- No prior knowledge about the characteristics of signals or noise is required → might perform badly at first.
- Chest signal (parent's ECG) is correlated with the noise (parent's ECG) in abdomen signal
  - The noise in abdomen signal can be removed, which reveals the wanted signal, which is the ECG of the fetus

# Noise cancellation using an adaptive filter

Abdomen ECG
(fetus + parent)




$x(n) = v(n) + m(n)$

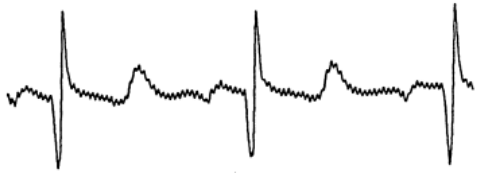


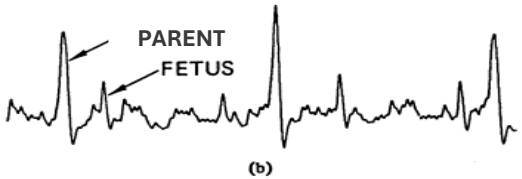

r(n)

Fetus ECG
(fetus)

Chest ECG
(parent)

$e(n) = x(n) - y(n) \approx v(n)$





# FIR filter

- Filtering is performed in a loop, N samples at a time through the whole signal
- For causal discrete-time FIR filter each value of the output is a weighted sum of the most recent input values:

$$
\begin{aligned}
y[n] &= b_0 x[n] + b_1 x[n-1] + \cdots + b_N x[n-N] \\
&= \sum_{i=0}^{N} b_i \cdot x[n-i],
\end{aligned}
$$

→ This is why we take **N most recent input data** of the parent signal (x)

where:

- $x[n]$ is the input signal,
- $y[n]$ is the output signal,
- $N$ is the filter order; an $N$th-order filter has $(N+1)$ terms on the right-hand side
- $b_i$ is the value of the impulse response at the $i$'th instant for $0 \leq i \leq N$ of an $N$th-order FIR filter. If the filter is a direct form FIR filter then $b_i$ is also a coefficient of the filter.

# Least mean square (LMS) algorithm

Objective: to obtain an output which is a least-square fit to the desired signal

1) Feed the output back to the adaptive filter and adjust the filter to minimize the total system output power
2) The system output is an error signal for the adaptive process

# The gradient LMS algorithm

1. Initialize the filter coefficients, e.g. as zeros
   - H = [0,0,0]. The size depends on N.
2. Select r(n) (most recent samples of parent chest signal*)
3. Calculate the filter output: $y(n) = \boldsymbol{H} * \boldsymbol{r_n}$
   - $H$ is column vector, $r$ is a column vector
   - In matrix multiplication (test yourself in MATLAB!):
   - $row * column$ = scalar, $column * row$ = matrix
4. Calculate the error signal : $e(n) = x(n_delayed) - y(n)$ **
5. Update the filter coefficients: $H = H + 2\mu e(n) r(n)$
   - Delta = 2μ
6. $n = n + 1$, go to step 2

r(n) = parent chest signal
y(n) = filtered parent chest signal
x(n) = abdomen signal (fetus + parent)
e(n) = error signal

* Select N most recent samples of parent_chest signal → 3rd, 2nd, 1st; 4th, 3rd, 2nd; 5th, 4th, 3rd;... So, the last, second last, third last (important!) The easiest is to make a variable r_n which contains the n most recent samples (selected again on each loop round). If r =[1,2,3,4,5] you can get choose samples in the reversed order by using the increment of -1 instead of 1. For example, r(5:-1:3) gives 5, 4, 3.

Check correct vector order: you need to transpose r_n at some of the calculations to have correct matrix multiplication. Thus, sometimes r_n is a row vector and sometimes a column vector.

** Remember to delay the abdomen signal x(n) by (N-1)/2.

# References

- The course book:
  - Chapter 3